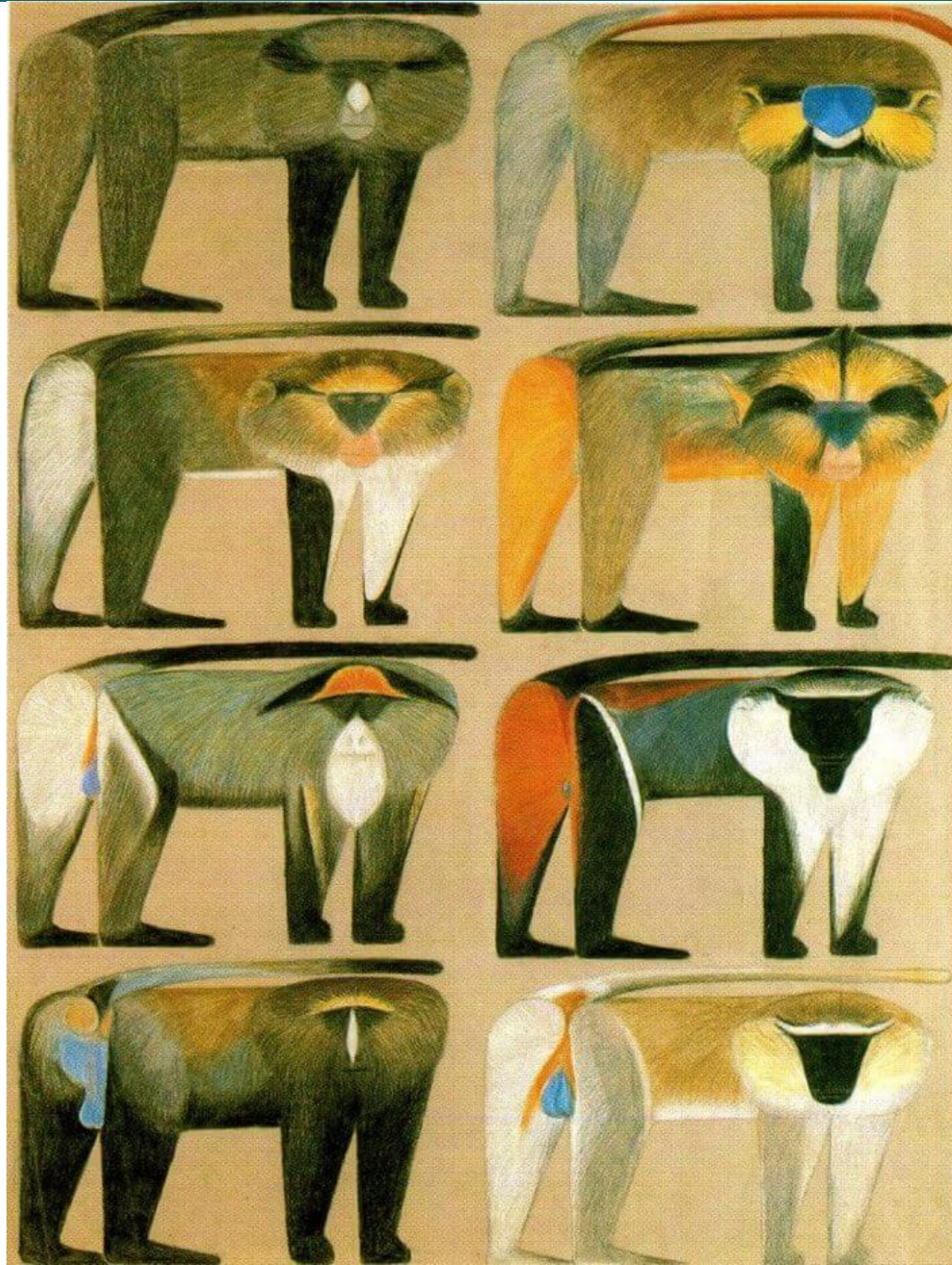




Culture and genetic variation

Dr. Laura van Holstein

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Room 215

Culture and genetic diversity

How does culture influence genetic diversity in modern humans? We've thought about gene-culture coevolution, of which our example was selection for lactase persistence in dairying communities, but how might human culture influence patterns of diversity within and between populations beyond this process?

PART 1: PREDICTIONS

In groups of 3-5, take five minutes to discuss the ways in which culture might influence the four microevolutionary drivers of diversity. Remember that a possible answer is: no effect at all!

Note down your predictions in the table on the next page.

One member of your group will briefly present your expectations to the class.

As the lecture progresses, note whether your expectations were met – or not!

□

DRIVER: Gene Flow

Possible effects of culture on this driver:

Migration as just how it works for humans

DRIVER: Selection

Possible effects of culture on this driver:

Disease resistance due to exposure from funerary & living ability, mate choice sexual selection

Increasing? Diversity by having positive selection for advantageous genes but lacks good deleterious

Processes discussed in the lecture:

Processes discussed in the lecture:

Were your expectations met?

Were your expectations met?

DRIVER: Drift

DRIVER:

Possible effects of culture on this driver: Possible effects of culture on this driver:

migration

Processes discussed in the lecture:

Processes discussed in the lecture:

Were your expectations met?

Were your expectations met?

□

PART 2: MARRIAGE

Patrilocality is a residential pattern in which newly married couple lives with or near the **husband's parents**; **matrilocality** is the opposite.

REVIEW FROM LECTURES 2-3: cross out incorrect answers.

- mtDNA is inherited through the **maternal line** / ~~through the paternal line / biparentally~~
- Y-chromosome DNA is inherited through the **maternal line** / **through the paternal line** / ~~biparentally~~

The Karen people are a matrilocality society. Take 5-10 minutes to discuss, in groups of 3-5, what figure best describes the expected pattern of genetic similarity for mtDNA and Y-chromosome DNA across 3 Karen villages distributed in a straight line. Then, answer the three questions below.

A)

mtDNA:



Y-chromosome:



B)

mtDNA:



Y-chromosome:



C)

mtDNA:

D)

mtDNA:

In each village, males come to this village from a bunch of different villages but families stay in their village



Y-chromosome:



Y-chromosome:



there would be more male diversity within villages

1. Under matrilocality, in **Village 1**, I expect:

- ☐ greater mtDNA diversity than Y-chromosome diversity
- ☒ lower mtDNA diversity than Y-chromosome diversity

2. Under matrilocality, if I **compare across villages**, I expect

- ☒ mtDNA diversity *within* Village 1 to be different to Village 3
- ☐ mtDNA diversity *within* Village 1 to be the same as Village 3

3. Under matrilocality, I expect

- ☒ mtDNA F_{ST} to be high
- ☐ mtDNA F_{ST} to be low

more variation between groups

large mtDNA diversity between groups

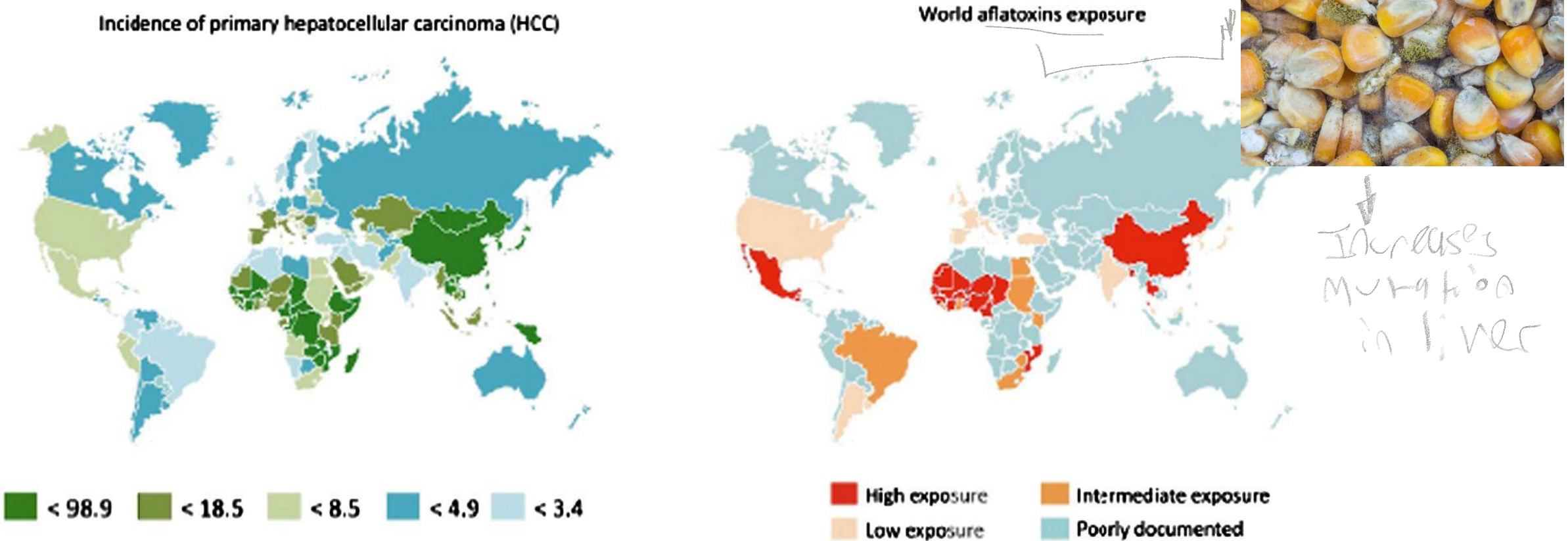
4. Under matrilocality, if I sample a single person from Village 1, Village 2, and Village 3, which samples will be the most similar to each other?

- ☒ mtDNA of Village 1 and 2
- ☐ mtDNA of Village 1 and 3

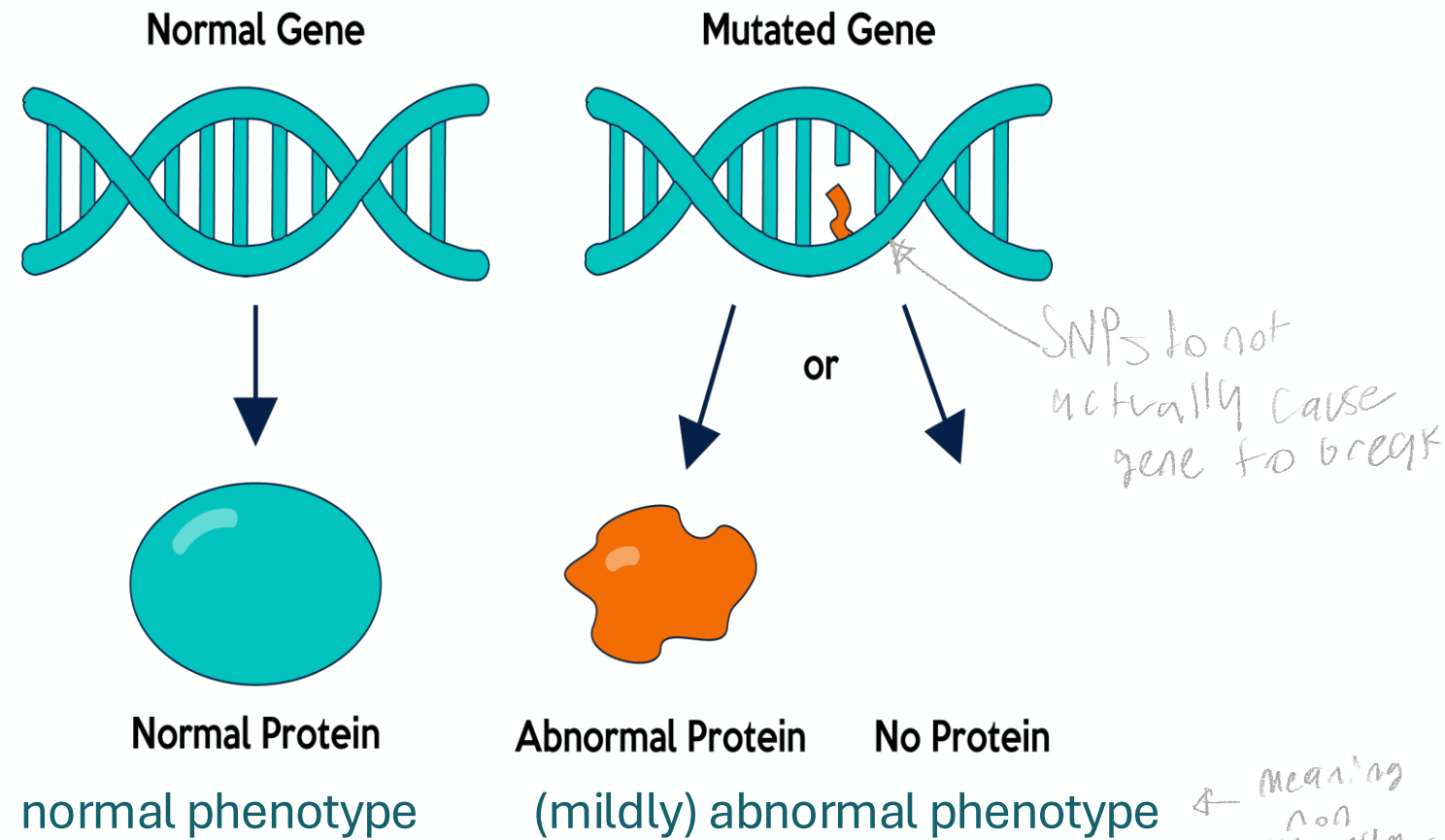
Which of these processes might be influenced by **culture** (or behaviour more generally)? **PART 1 of your worksheet**



Behaviour can lead to mutagen exposure



...but limited evidence for effects in the germline (in humans).



Unbuffered environment:
Mutation selected out

Buffered environment: + fewer selective pressures
Mutation persists

We will only be thinking of deleterious variants for this



Does parental care cause mutations to accumulate?

wild-caught individuals



experimental evolution

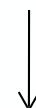
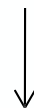
2 full care lines

2 no care lines



20

generations



assessed deleterious mutations in
each population by:

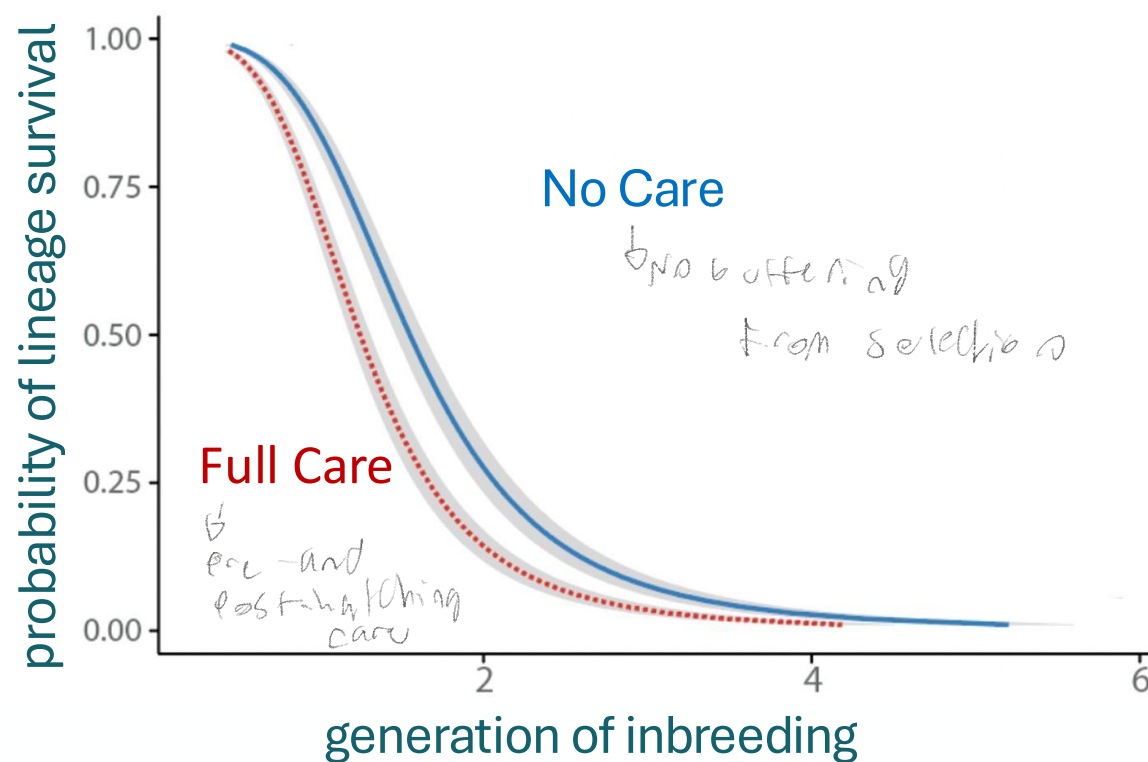
- a) sequencing (Mashoodh et al. 2023)
- b) inbreeding (Pascoal et al. 2023)

provide food for
their offspring
and guard them



Inbreeding reveals higher mutation load in 'Full Cares'

Survival of inbred lineage after 20 generations:

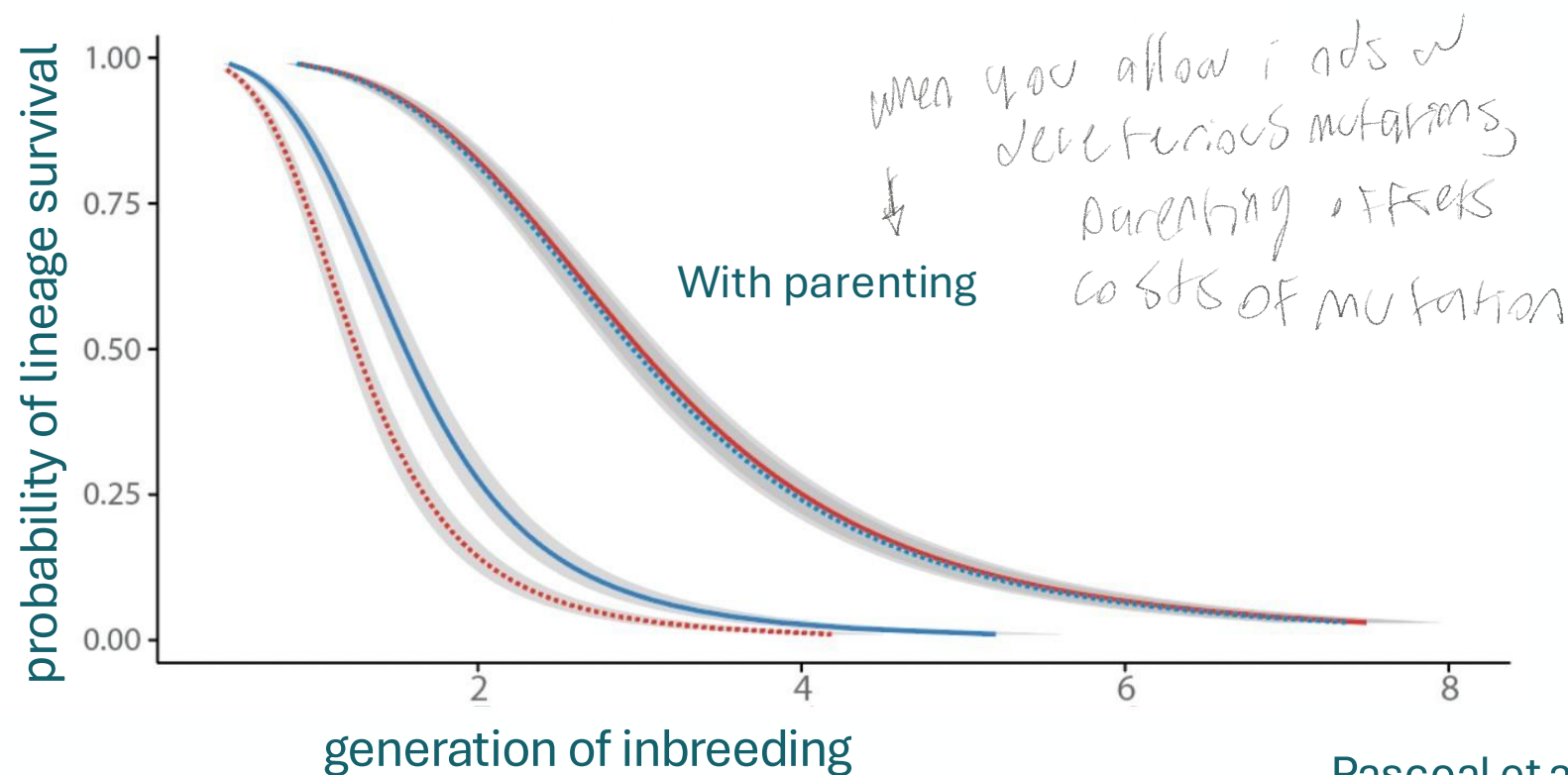


Pascoal et al. (2023)



Inbreeding reveals higher mutation load in 'Full Cares' ...but the cost of these mutations is offset by parenting!

Survival of inbred lineage after 20 generations:

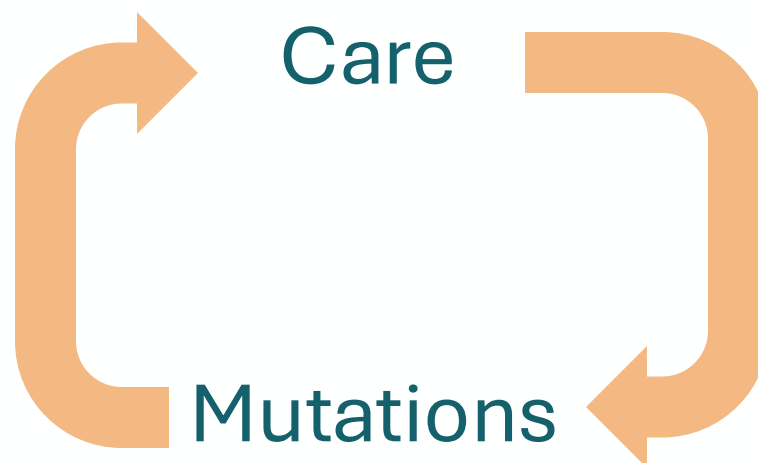


Pascoal et al. (2023)



Is care self-reinforcing?

- Care causes mildly deleterious mutations to accumulate...
- ...but care is the phenotypic antidote for an increased mutation load



Pascoal et al. (2023)

Which **cultural** processes might influence drift, gene flow, and/or selection?

There are more,

Marriage practices

Inheritance of
power and property

Language and
identity

Warfare

Beauty standards

Matri/Patrilocality: Male vs female migration network

Endogamy/exogamy: isolation and local drift (e.g. Indian castes) vs more mixing

Group identity: Rate of admixture on contact between groups.

Polygyny/polyandry: reproductive skews (more skew -> more drift)

Power skews: Culturally inherited reproductive success (drift)

Dispersal tendencies: When are villages/bands founded/go extinct? When do groups/individuals migrate?

Population densities/sizes: families vs bands vs villages vs cities

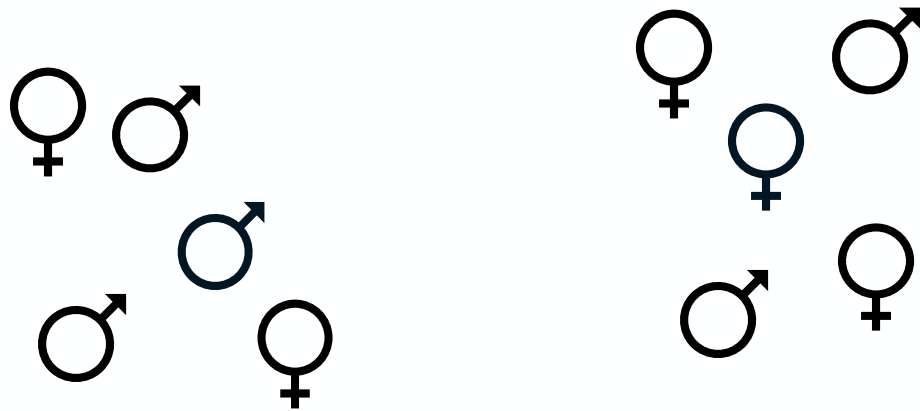
Assortative/disassortative mating

Marriage traditions leave distinct signals in the genome... but not all of the genome!

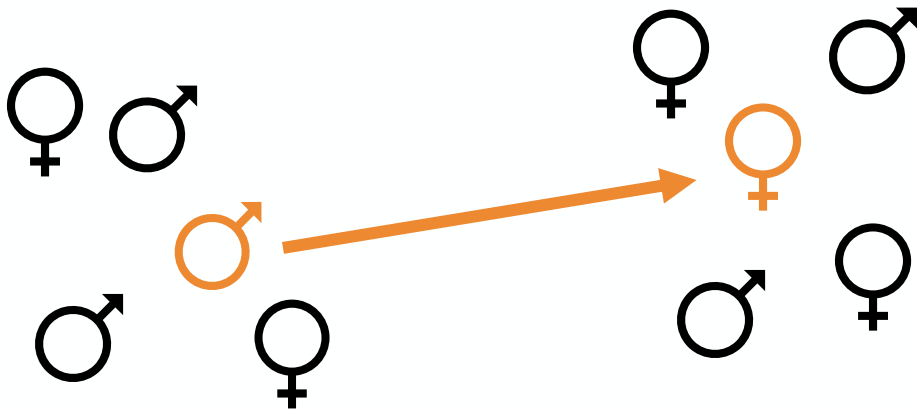
Marriage practices

← what we will be focusing on today





Mafar loc ality



Mae: locality



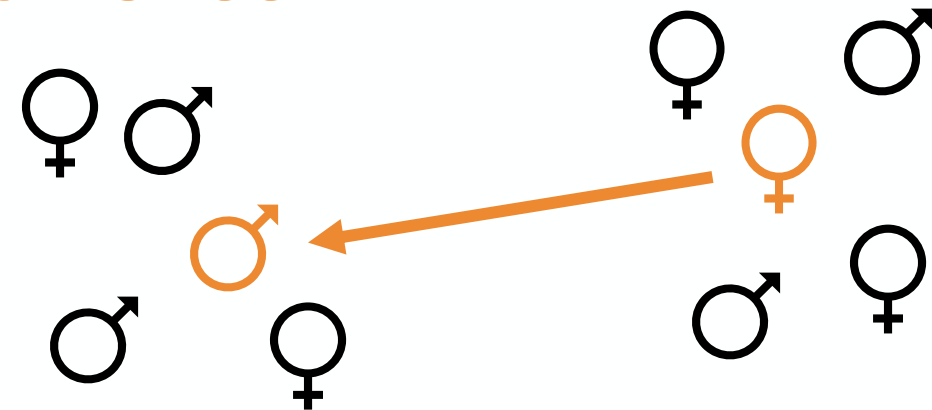
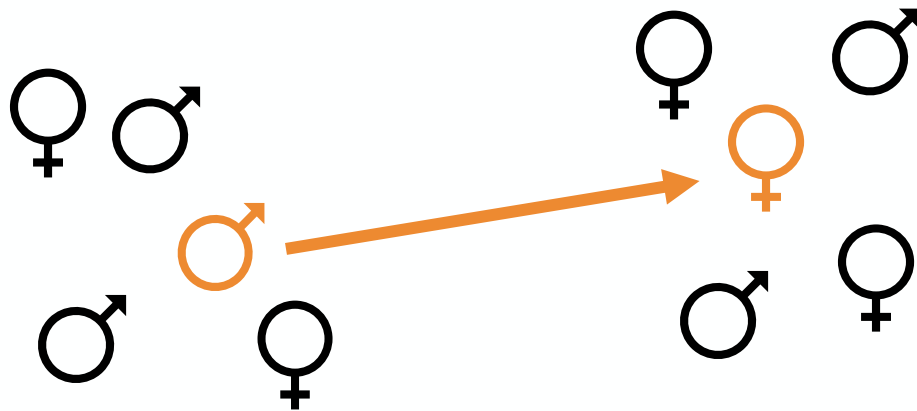
Karen

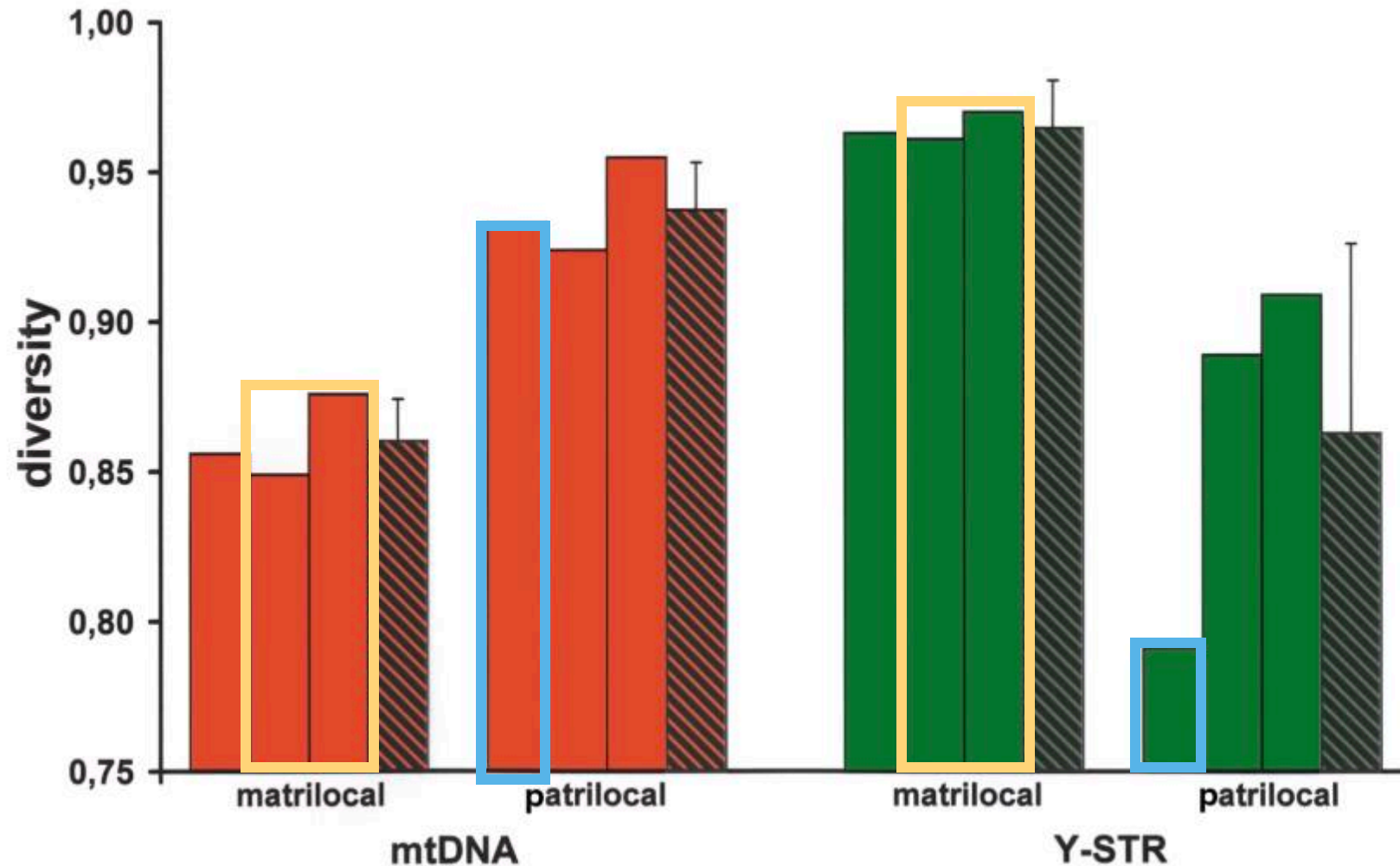
Parai locality



Akhu

PART 2 of your worksheet





Matrilocal: expect greater Y-chromosome diversity

- mtDNA less diverse
- Y more diverse



Patrilocal: expect greater mtDNA diversity

- mtDNA more diverse
- Y less diverse

What is going on?

Key

(a)

Female

Male



A A



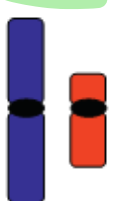
X X



mtDNA



A A



X Y



mtDNA

Neutral Expectations

$$\frac{\pi_X}{\pi_A} = \frac{3}{4} = 0.75$$

$$\frac{\pi_Y}{\pi_A} = \frac{1}{4} = 0.25$$

$$\frac{\pi_{mt}}{\pi_A} = \frac{1}{4} = 0.25$$

Under random mating and neutrality, diversity of autosomes/ X/Y/mtDNA follow the **ploidy of transmissible copies** in the population

Under random mating with no selection

Neutral expectations:

- **N_e or diversity follows a 4:3:1:1 ratio across the autosomes, X, Y, and mtDNA respectively**
 - Males have 1 x chr, females have 2*
 - Everyone has the same autosomal amt of DNA*
 - That is, autosomal diversity is expected to be 4x higher than mtDNA diversity; and Y and mtDNA are expected to be equally diverse
 - only 1 sex for Y*

Sex-biased processes drive divergence from these expectations.

- Thus, comparing diversity of loci can provide a view into male vs female demography.

What is going on?

(a)

Female



A A



X X



mtDNA

Male



A A



X Y



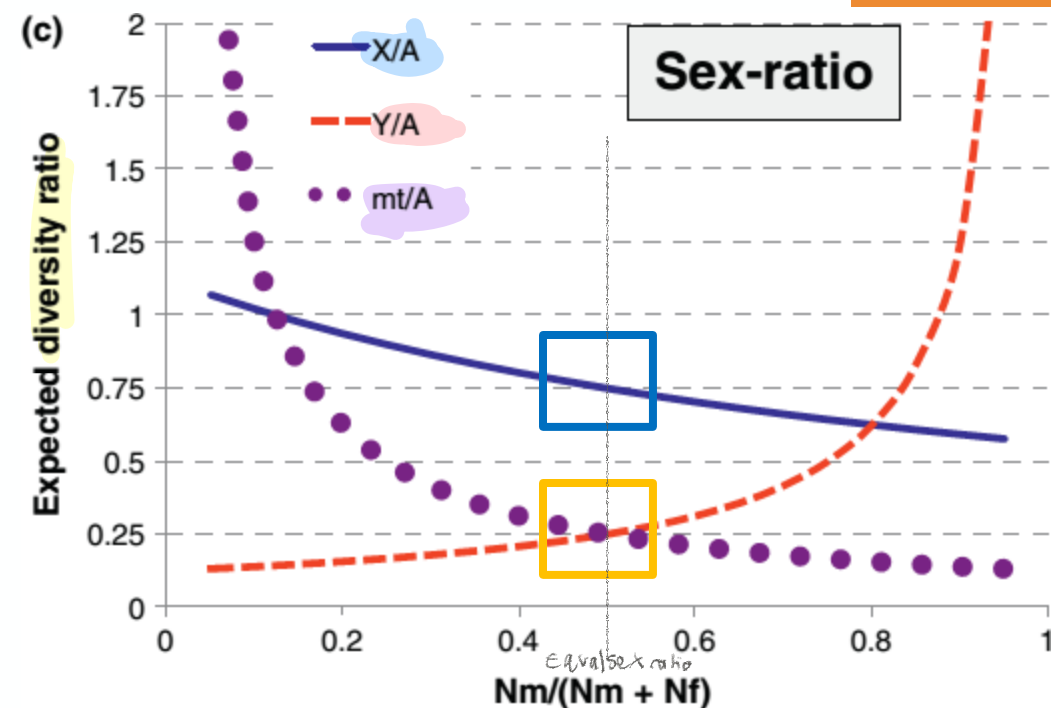
mtDNA

Neutral
Expectations

$$\frac{\pi_X}{\pi_A} = \frac{3}{4} = 0.75$$

$$\frac{\pi_Y}{\pi_A} = \frac{1}{4} = 0.25$$

$$\frac{\pi_{mt}}{\pi_A} = \frac{1}{4} = 0.25$$



What is going on?

(a)

Female



A A



X X



mtDNA

Male



A A



X Y



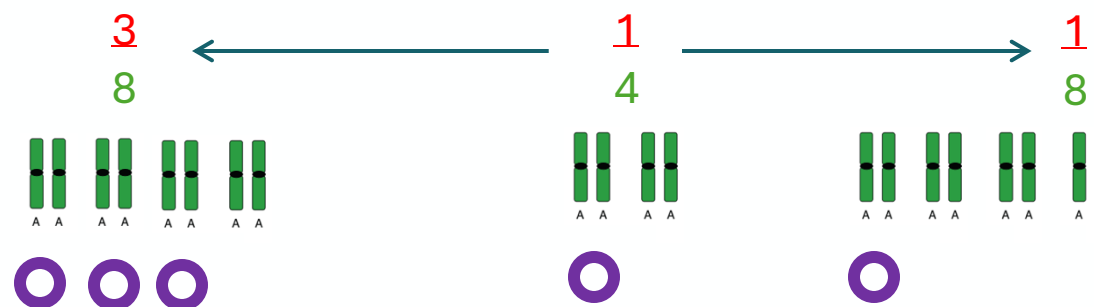
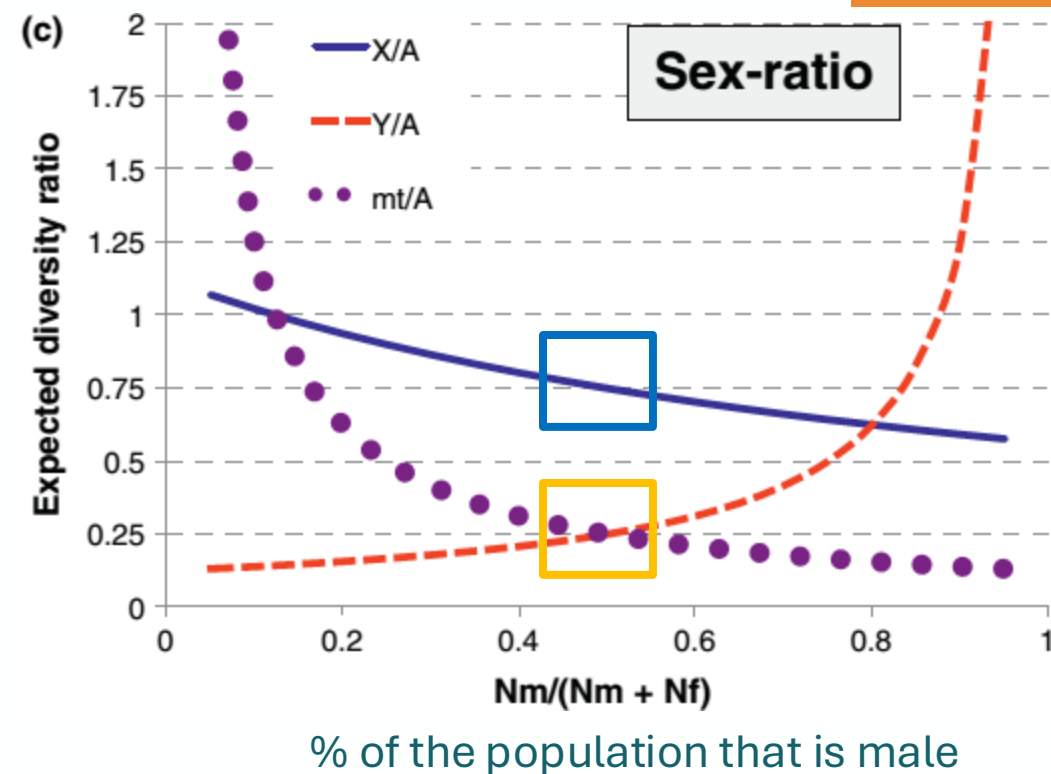
mtDNA

Neutral Expectations

$$\frac{\pi_X}{\pi_A} = \frac{3}{4} = 0.75$$

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$$\frac{\pi_{mt}}{\pi_A} = \frac{1}{4} = 0.25$$



Key

What is going on?

Processes that can drive these shifts:

- Differential reproductive success
- Generation time
- Migration
- Sex ratios



Marriage practices

Key

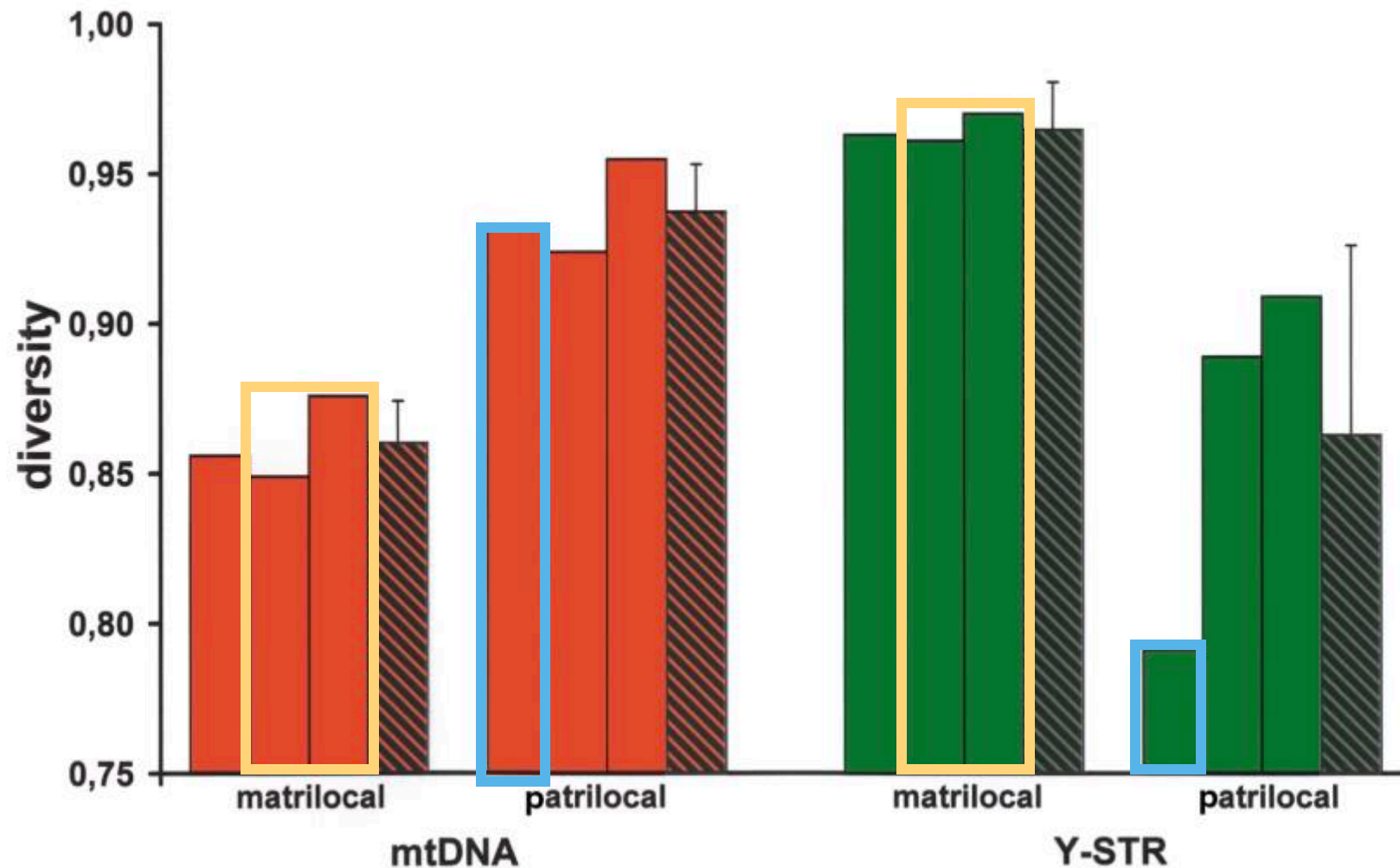
What is going on?

Processes that can drive these shifts:

- Differential reproductive success
- Generation time
- **Migration**
- Sex ratios



Marriage practices



Matrilocal: expect greater Y-chromosome diversity

- mtDNA less diverse
- Y more diverse

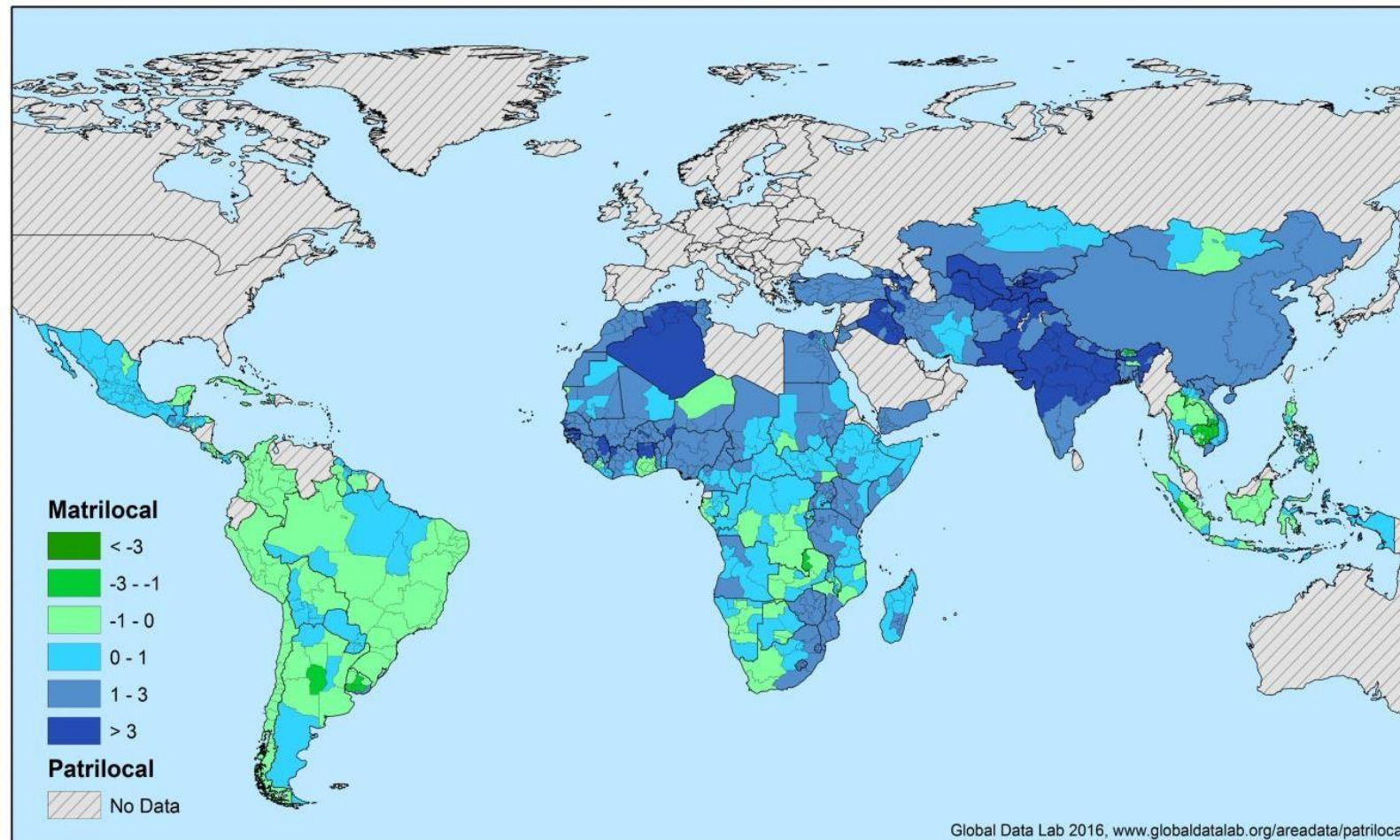


Patrilocal: expect greater mtDNA diversity

- mtDNA more diverse
- Y less diverse

Do the same patterns hold at a global scale?

Patrilocality vs Matrilocality of Married Couples in Subnational Regions of LMICs



**Matrilocal: expect greater
Y-chromosome diversity
*at small scales!***

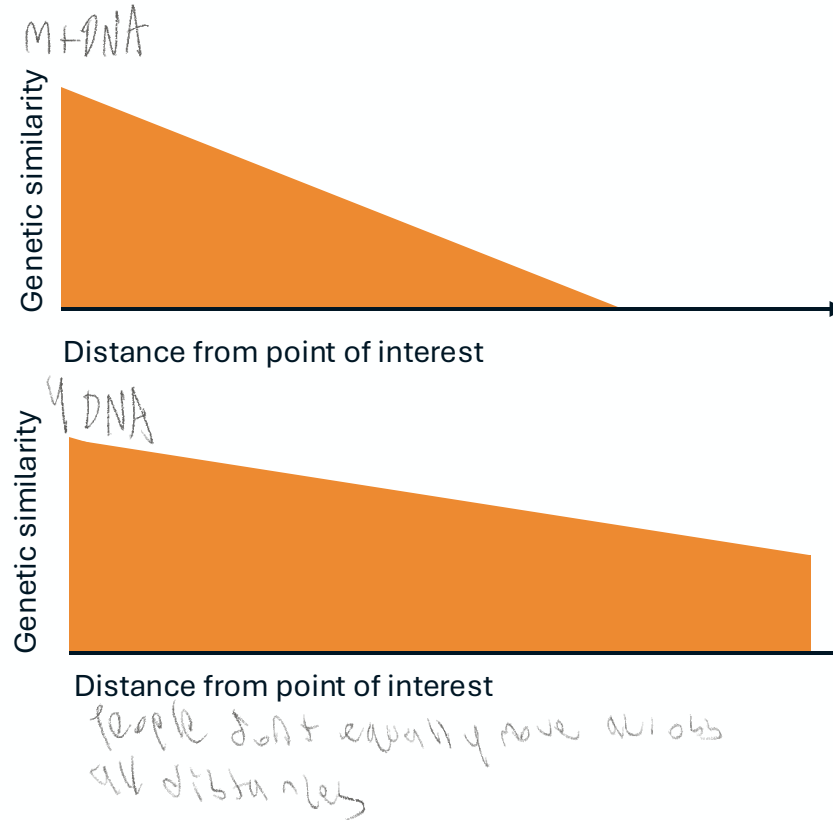
**Patrilocal: expect greater
mtDNA diversity
*at small scales!***

Do the same patterns hold at a global scale?

mtDNA:



Y-chromosome:



Matrilocal: expect greater Y-chromosome diversity at small scales!

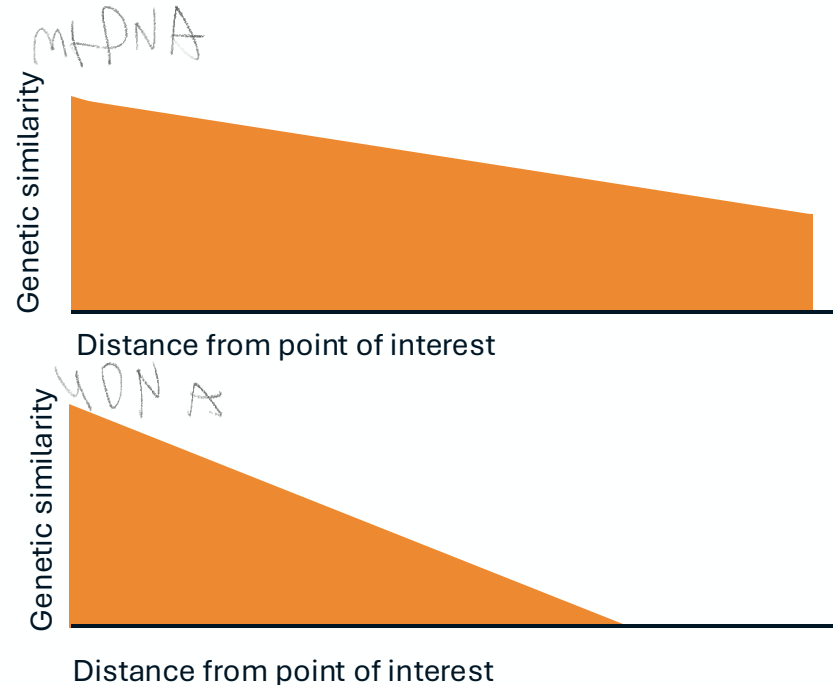
Patrilocal: expect greater mtDNA diversity at small scales!

Do the same patterns hold at a global scale?

mtDNA:



Y-chromosome:

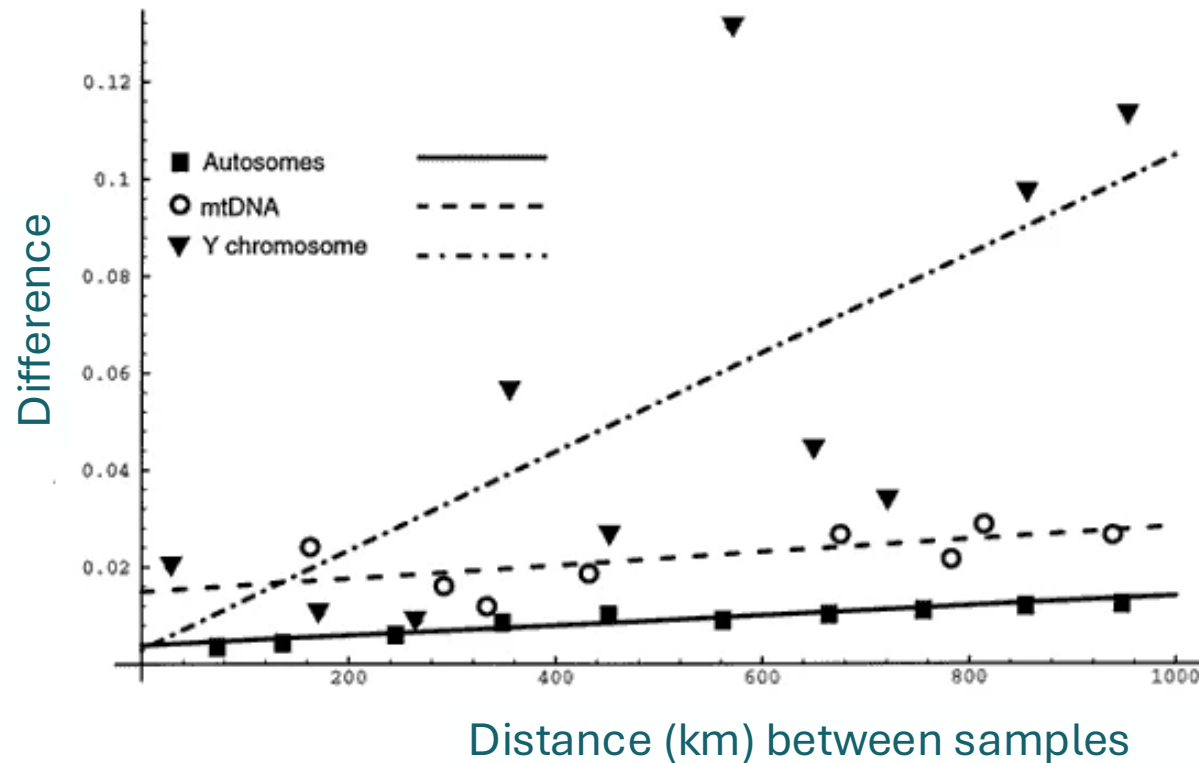


Matrilocal: expect greater Y-chromosome diversity *at small scales!*

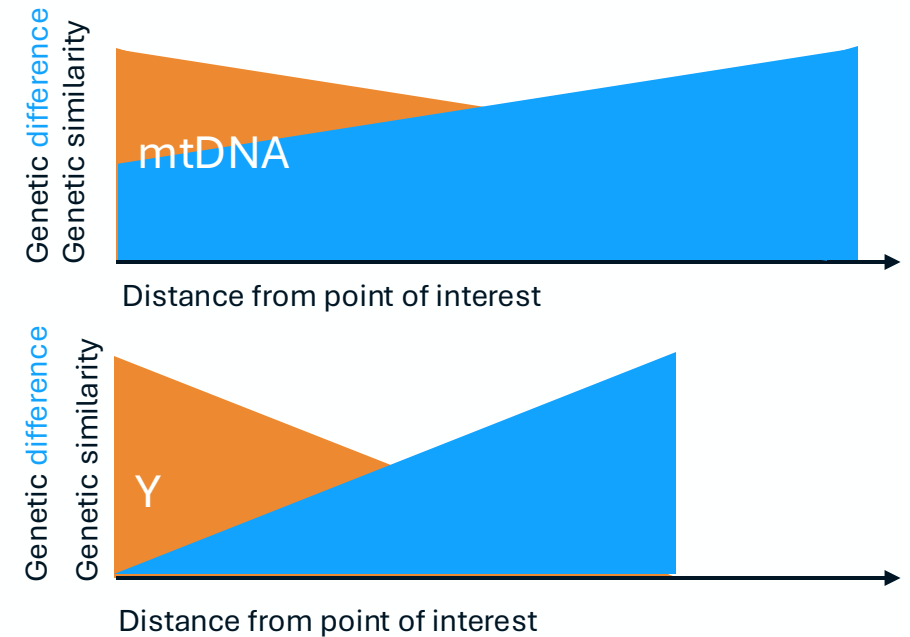
Patrilocal: expect greater mtDNA diversity *at small scales!*

Do the same patterns hold at a global scale?

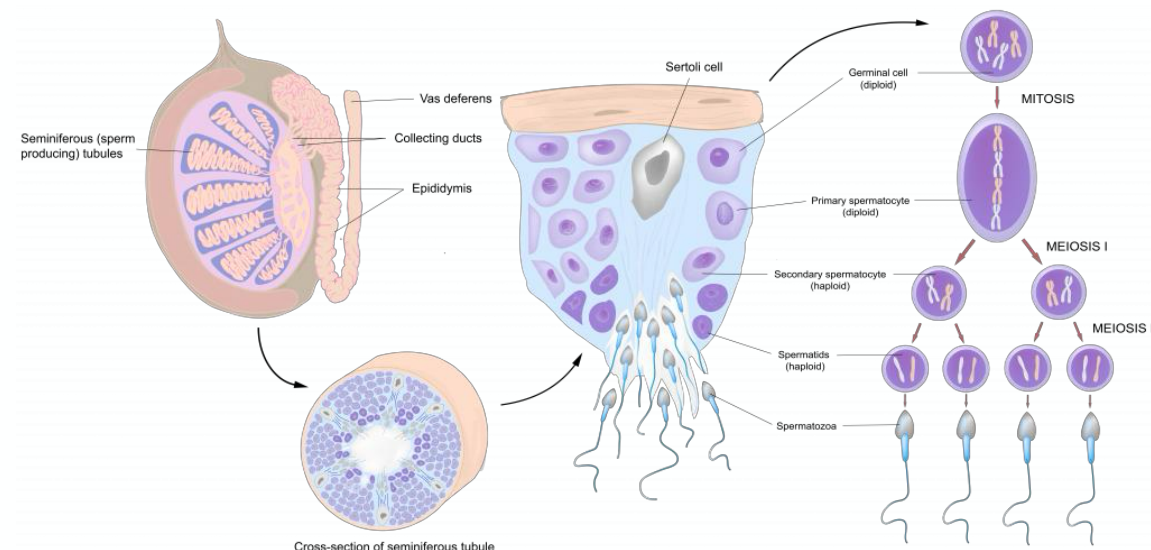
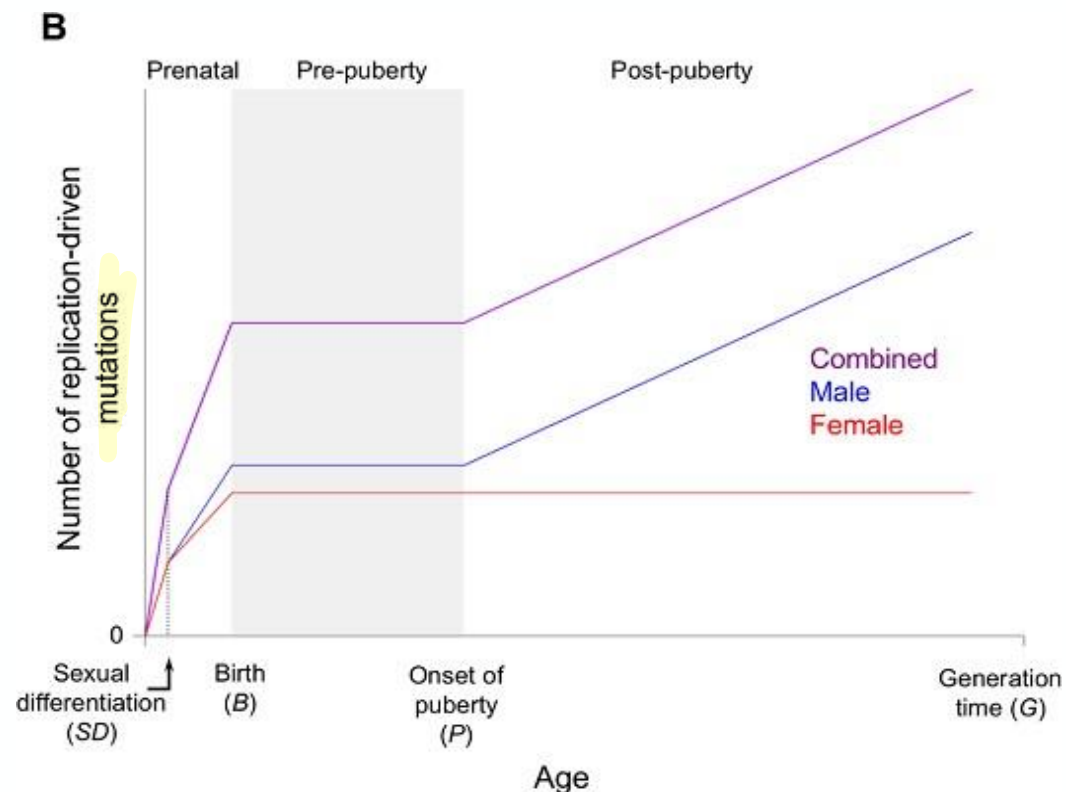
Patrilocality + more prevalent globally



Low mtDNA and high Y-chromosome differences
between groups, concordant with patrilocality



Biological drivers of different diversity patterns on mtDNA, X- and Y-chromosomes

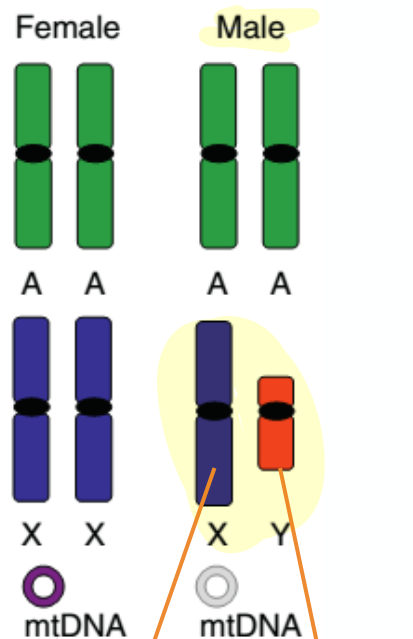


Male mutation bias

- Greater number of germline cell divisions in males than in females
- Sperm produced throughout life, and requires mitosis
- Increases diversity in this order: Y chromosome >> autosomes > X chromosome

Biological drivers of different diversity patterns on mtDNA, X- and Y-chromosomes

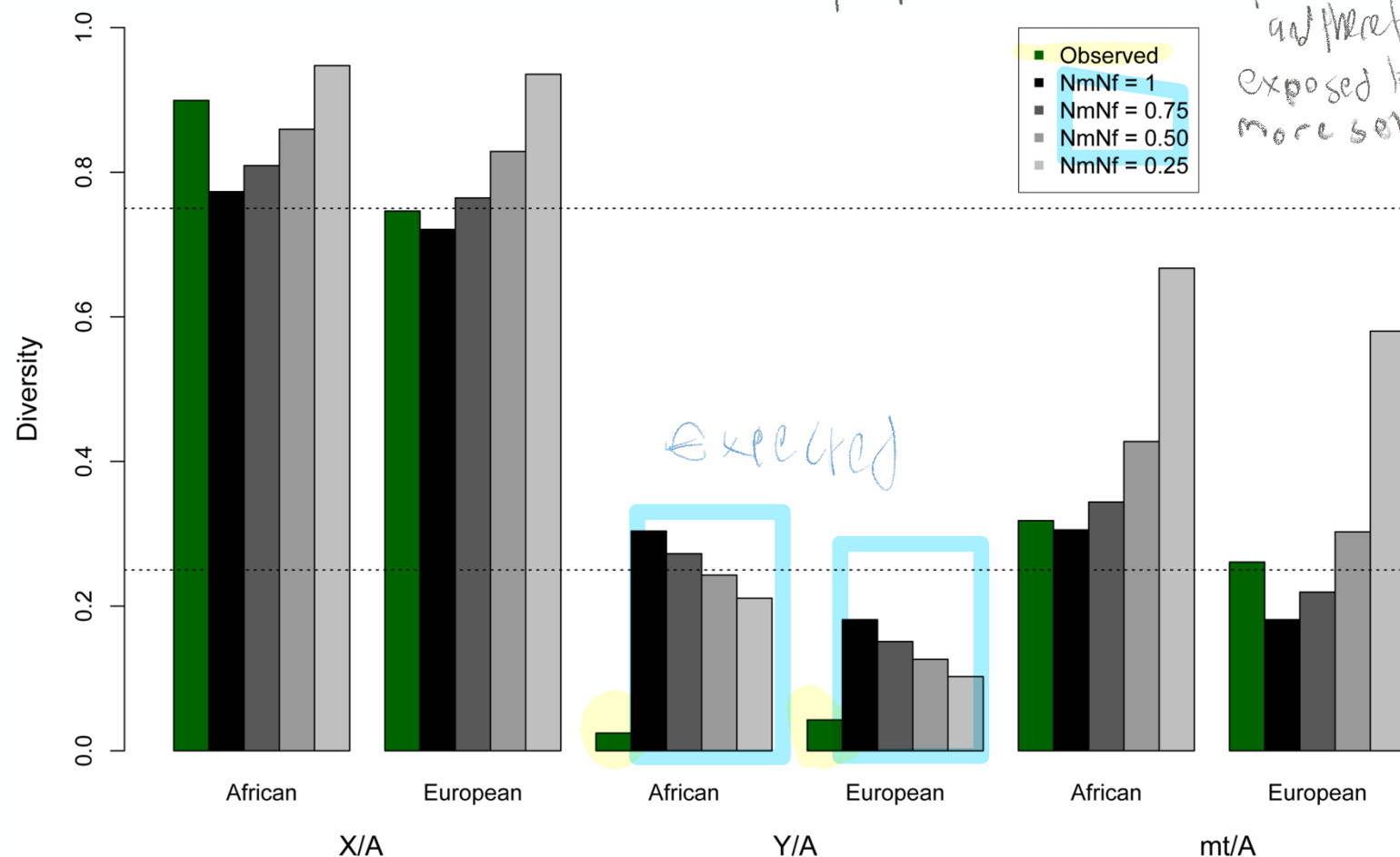
(a)



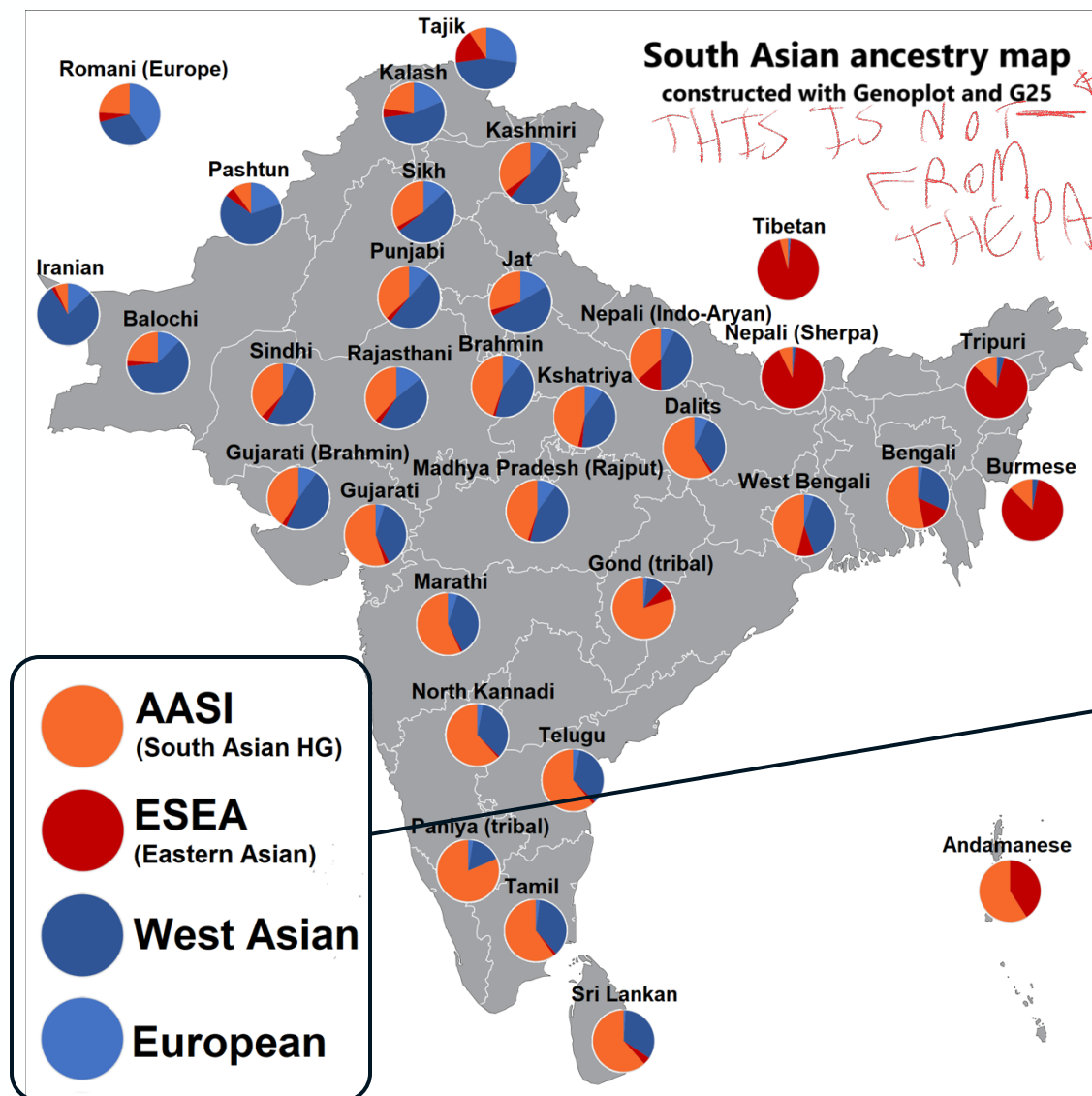
Recessive alleles
directly exposed to
selection

Purifying selection on the Y-chromosome led to reduced diversity on the Y-chromosome relative to X & mtDNA

only 1 ychr so it is inherently expressed and therefore exposed to more selection



The curious case of India



If mixing persists...

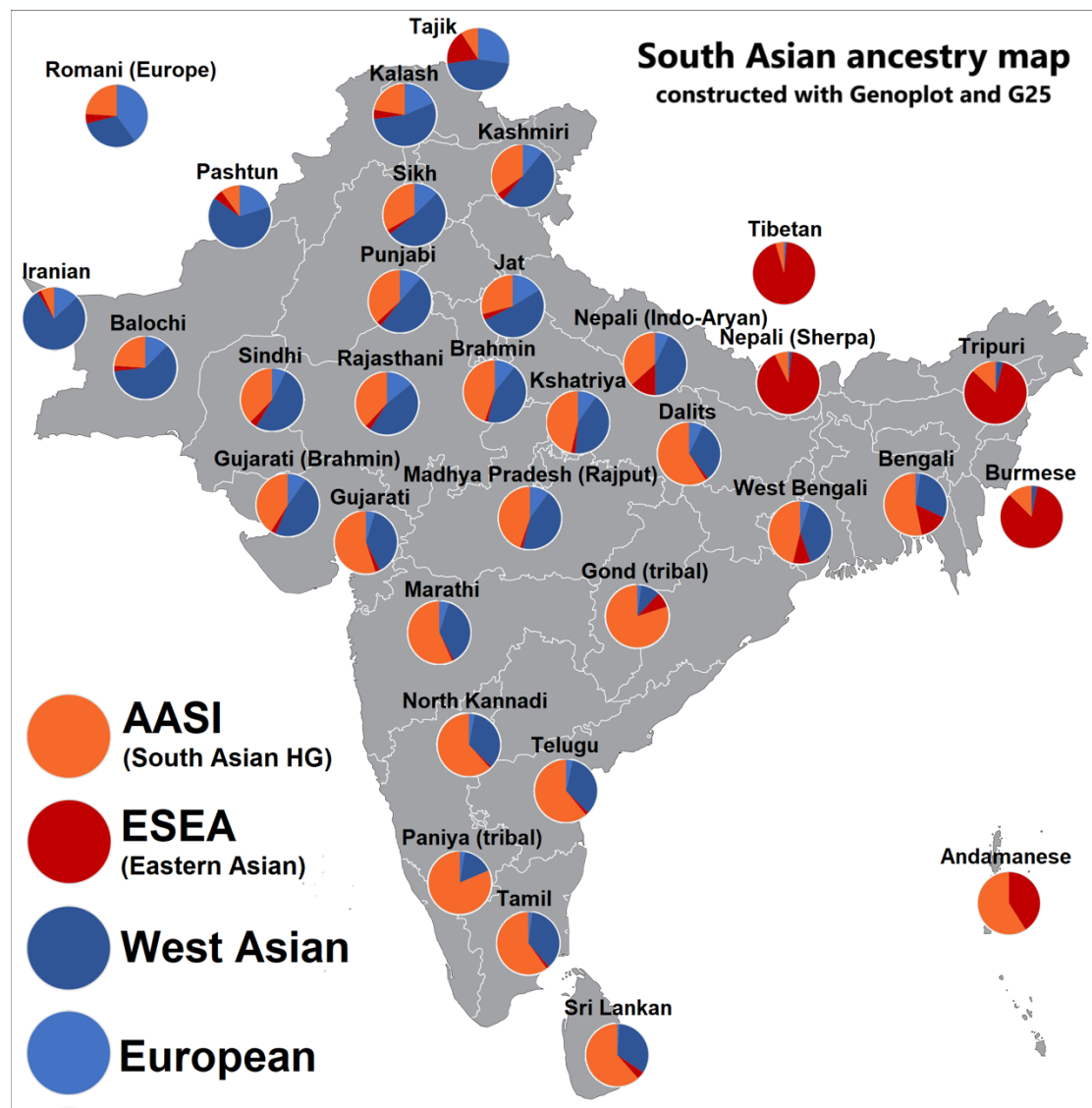


Amalgamation of all ancestries

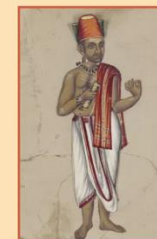
Mixing stops 1900 years ago. Why?

Castes

The curious cas(t)e of India



THE VARNA SYSTEM OF INDIA



ASTRONOMER

BRAHMIN

The highest-ranked of the four *varnas*, or traditional social classes of India. Includes Hindu priests, advisers, and intellectual leaders.



CHIEF

KSHATRIYA

The second-highest of the *varnas*. Includes rulers, military leaders, and large landowners.



ACCOUNTANT

VAISHYA

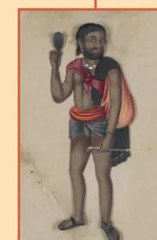
The third-highest of the four *varnas*. Includes merchants, traders, and farmers.



POTTER

SHUDRA

The lowest-ranked varna. Traditionally includes artisans, laborers, and servants.



BEGGAR

DALIT

(formerly called "untouchable")

A fifth category, with no *varna* designation. Includes various low-status persons and those outside the caste system.

Image credit: Beinecke Rare Book and Manuscript Library, Yale University

© Encyclopædia Britannica, Inc.

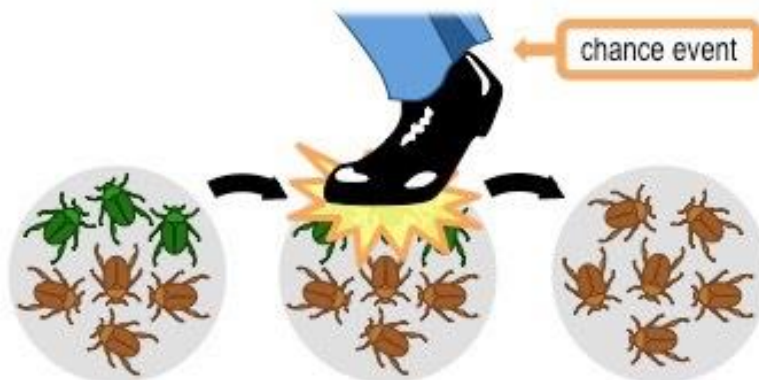
Which of these processes might be influenced by **culture** (or behaviour more generally)?

- Behaviour can, in some cases, influence mutation *rate* (mutagen exposure)
- But it likely has the largest effect on controlling whether mutations are exposed to selection

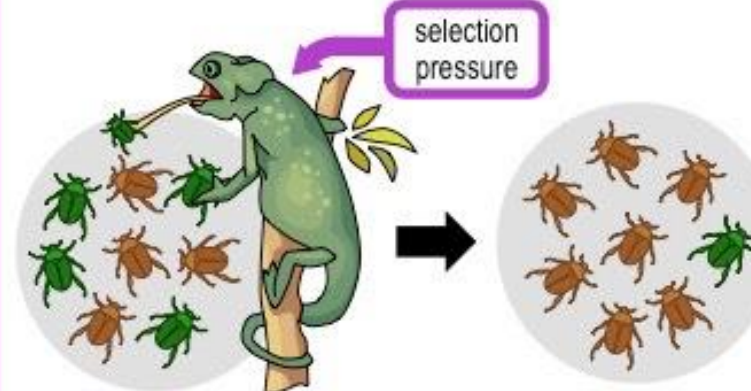
Mutation

- Culturally prescribed marriage practices affect mtDNA and Y-chromosome diversity
- ...and whole-genome signals!

Gene Flow



Genetic Drift



Selection

What is going on?

Processes that can drive these shifts:

- Differential reproductive success
- Generation time
- Migration
- Sex ratios



Marriage practices

What is going on?

Processes that can drive these shifts:

- **Differential reproductive success**
- Generation time
- Migration
- Sex ratios



Marriage practices

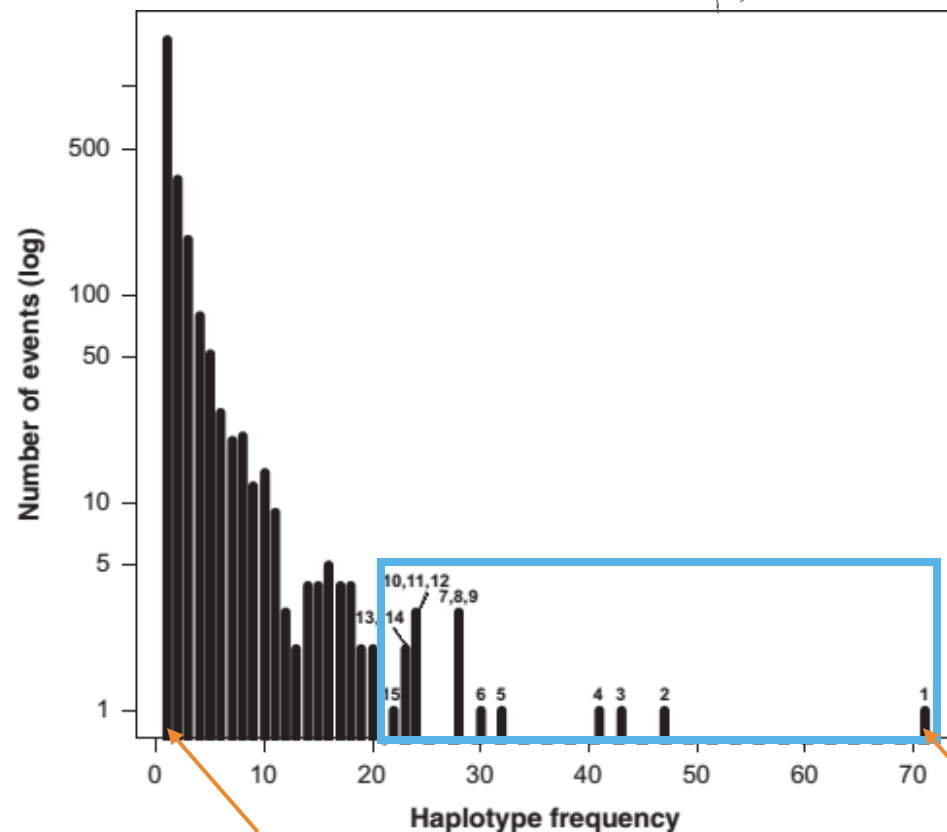
1) Asian Y-chromosome haplotypes

15 haplotypes out of 2552 represent ~40% of genetic data

- Differential reproductive success

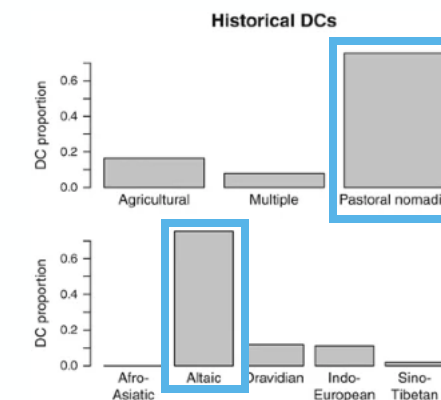
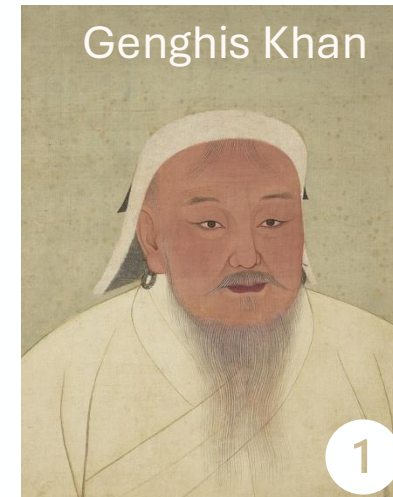
Due to stratified society?

There have only been a few Hchs that are being passed down a lot of



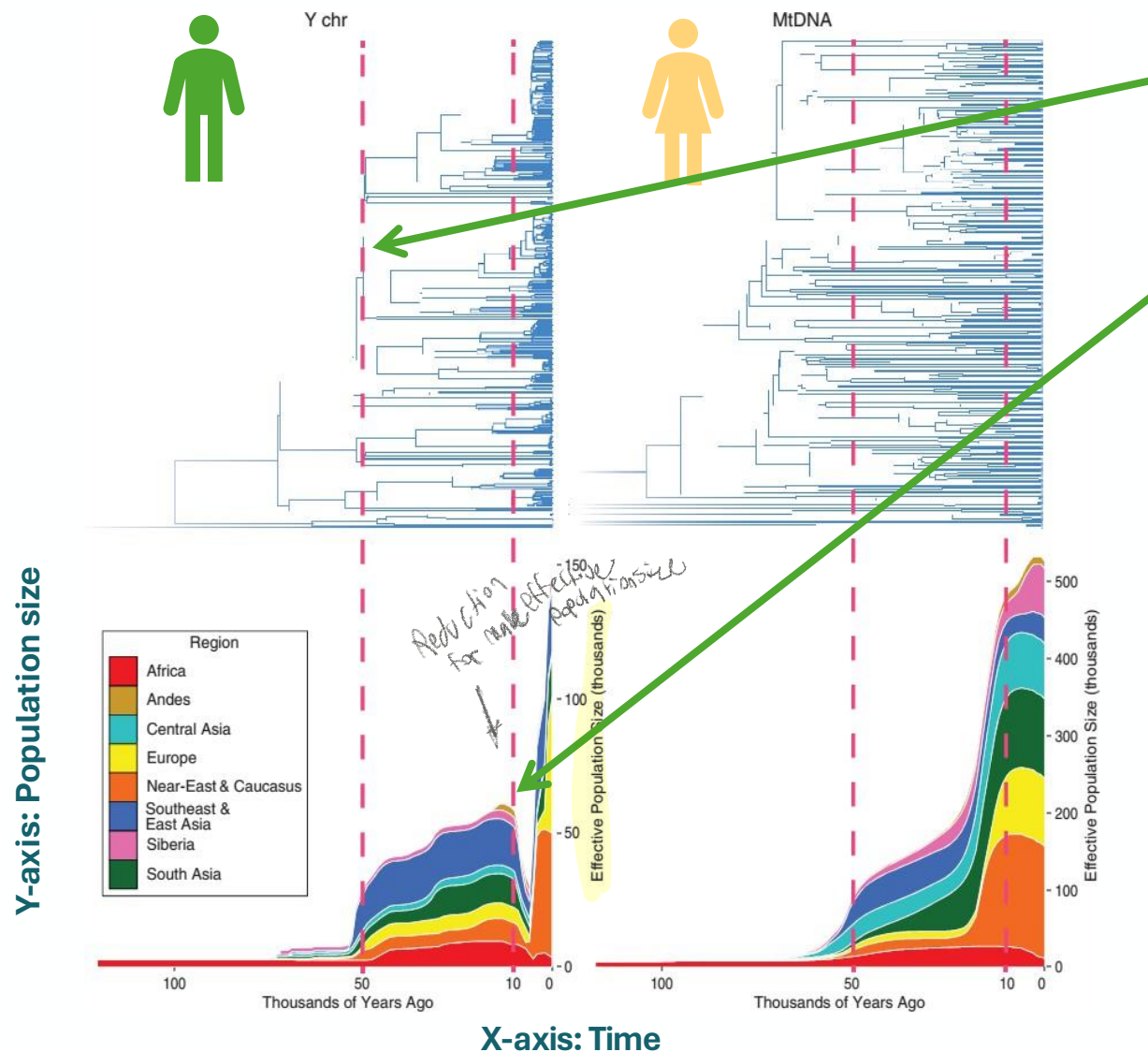
>500 haplotypes occur only once

1 haplotype occurs 70 times



Pastoral nomadism, Altaic speaking

2) A male-specific population collapse in the Neolithic?



Coalescence of non-African populations

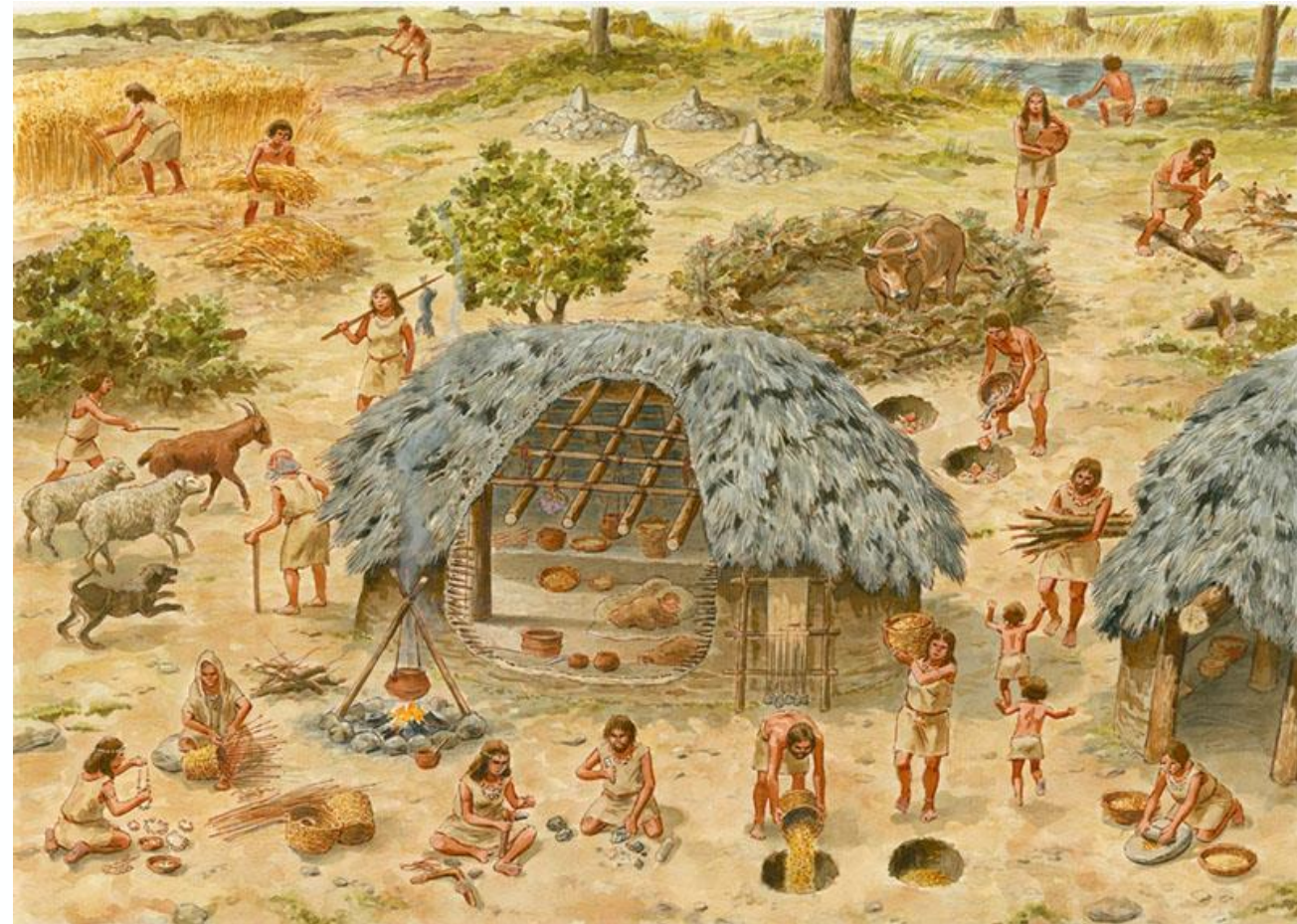
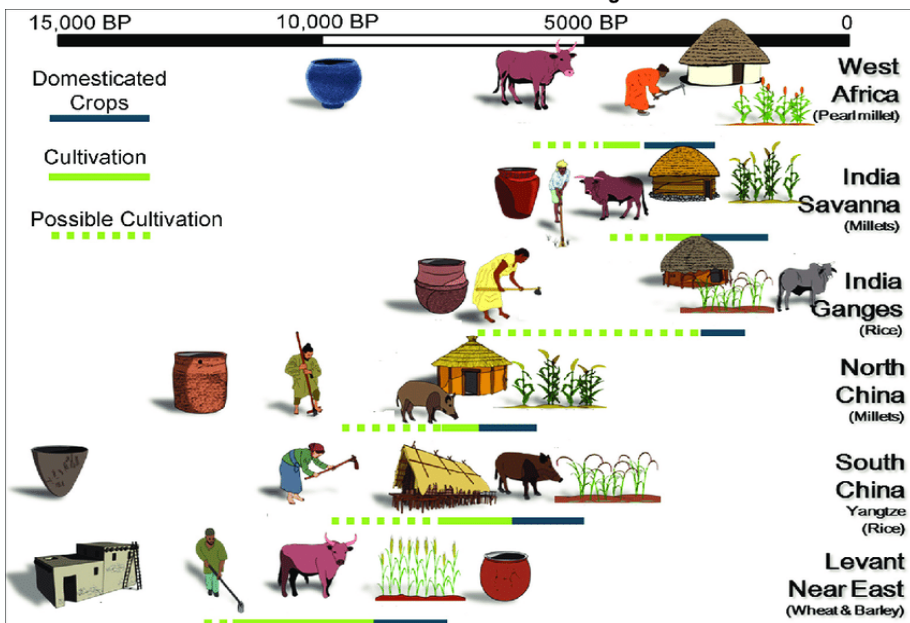
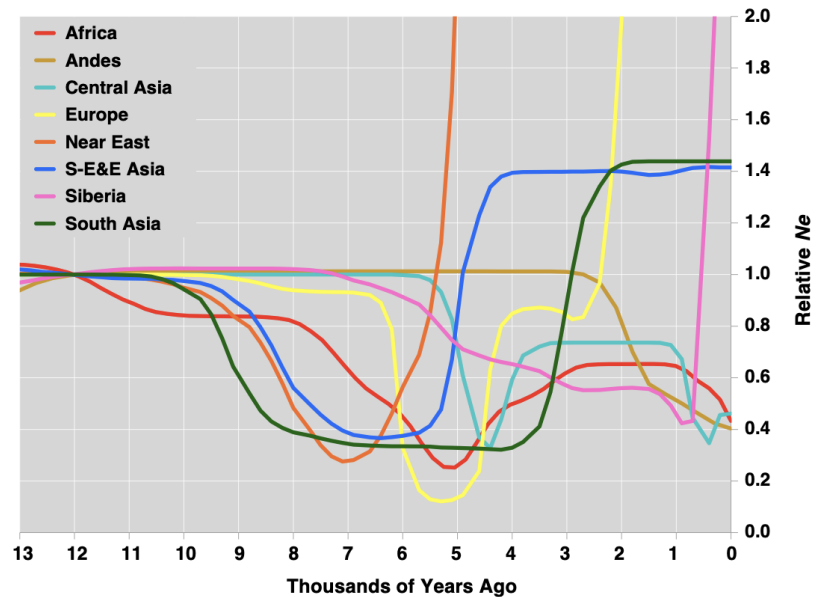


Archaeological evidence for OOA **50 kya**

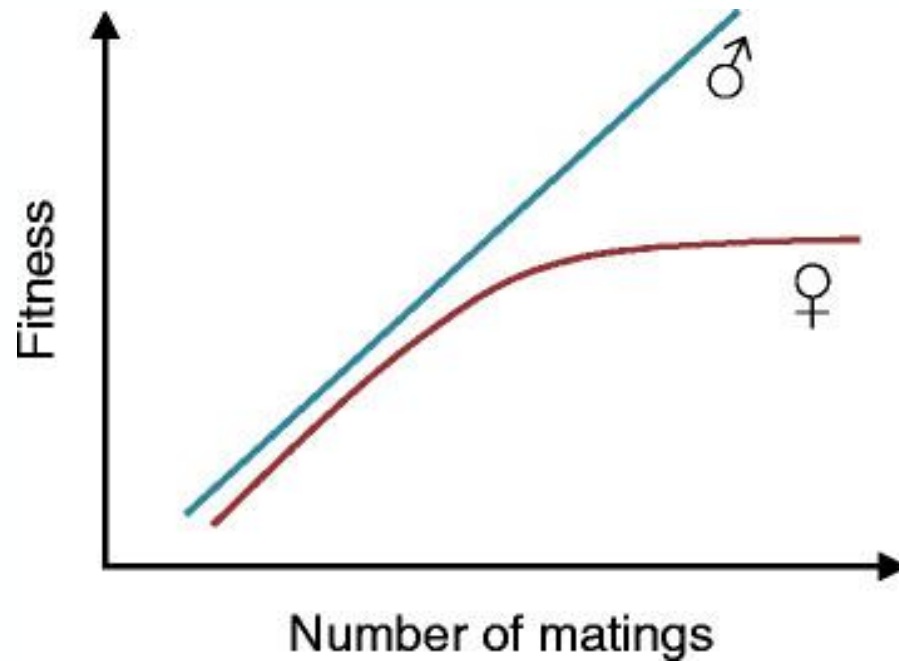
10 kya: steep drop in male effective population size

- Selection on Y-chromosome?
- Changes in variance of reproductive success?

A male-specific population collapse in the Neolithic?



Why not females (as much)?



Patrilocality and variance in male reproductive success are not uncorrelated!